

AIS Glass — Data Validation and Forecast Report

Demand Forecast & Inventory Intelligence — Proof of Concept

Prepared Oct 6, 2026

What this report is for

AIS Consumer Glass Solutions gave us five files covering their order history, their sales, their stock position, their product list and their branch list. We built a system on top of that data which shows what each branch ordered, what it actually received, where it fell short, and what it is likely to need next.

The obvious question is whether the numbers on our screens are the same numbers that are in their files. This report answers that question.

They are the same. Every quantity matches exactly — across all 53 branches, all 2,063 products, every product category and every month. On rupee value, 51 of the 53 branches match to the rupee. The remaining difference is ₹1,66,548 out of ₹857 crore, and we have traced it to two specific order lines, which are explained later in this report.

Getting to that point took three rounds of correction. We found three genuine mistakes in our own calculations along the way, and we have written all three down rather than quietly fixing them. Two further things looked like mistakes but turned out to be errors in our checking rather than in the system, and those are written down too.

The rest of this report covers what is in the data, what the data says about how the business is running, how we checked it, and what we found.

The information AIS gave us

Five files in total. We treat all five as read-only — nothing we do writes back to them, and the copies we work from are untouched.

File	Size	Sheet inside it	Rows
Orders & Receipts (Lead Time)	64 MB	Receipts (Lead Time)	7,75,912
Sales Data FY 24~26	79 MB	FY 25-26 Sales	17,03,042 across both sheets

File	Size	Sheet inside it	Rows
		FY 24-25 Sales	
Stock in Hand as on 1st Aug'26	6.7 MB	Closing Stock Jul'26	1,44,921
Substitution Mapping	218 KB	Sheet1	2,417
Location Master	24 KB	single sheet	57

What is inside each sheet

The orders sheet is the most important one. Every row is a single order line — one branch asking for one product on one date. It records the branch, the product, the date the order was placed, the date it was despatched, how many units were asked for, how many were actually sent, and the price. Because it holds both the request and what was supplied, it is the only file that can tell us where the business is falling short. Everything else in this report is built around it.

The two sales sheets record what was invoiced. One covers 2025-26, the other 2024-25, and between them they hold just over 17 lakh lines. An invoice tells us what was billed, not what was asked for — so if a branch wanted 100 pieces and only 60 were available, the invoice says 60 and the shortfall is invisible. We therefore use these sheets only to see further back in time, not to measure service. These sheets also carry customer names, GST numbers and consignee details. All of that is removed the moment the file is read and none of it appears anywhere in the system.

The stock sheet is a single photograph, not a film. It lists what was sitting in each branch on 1 August 2026 — the quantity, its value, and when it was last received. There is no record of what stock was on any earlier date. This matters: we can advise on today's position, but we cannot test how a stocking policy would have performed last year, because there is no history to test it against. The first row of this sheet is a heading banner rather than data, so the real column names sit on the second row.

The substitution file is the product catalogue. For each of 2,417 products it gives the name, up to two alternative products that can be fitted instead, current prices, and the groupings every product chart uses — glass type, vehicle type, vehicle age band, whether it is an original-equipment part, and an A/B/C/D importance class.

The location file is the branch directory. Fifty-seven branch records with region, zone, supplying hub, tier, and the planning figures for each branch — transit time, average lead time, variation in lead time, service factor and truck load size. Like the sales sheets it

carries sensitive detail — PAN, GSTIN, contact names, phone numbers and addresses — and all of that is dropped on reading. Its first row is also a banner rather than data.

Two things in the files worth knowing about

The three replenishment columns in the branch directory are empty. They are zero for all 57 branches. We therefore never use them as a minimum order size — if we did, every reorder in the system would come out as nil.

Some lead time entries are not usable. The raw field contains entries of 0000-00-00 and some values that work out as negative time. We calculate lead time only from dates that make sense, and we report how many were excluded rather than hiding them.

What the order book actually says

Before getting to the checking, it is worth setting out what the data shows, because several of the findings are more interesting than the reconciliation itself.

The order history runs from **1 April 2025 to 31 July 2026** — sixteen months, 7,75,912 order lines, 53 branches, 2,063 different products.

The headline position

	Units
Ordered by branches	26,02,392
Actually despatched	21,33,961
Served	82.00%

In other words, **roughly one unit in every five that a branch asked for did not arrive.** Over sixteen months that is 4,68,431 units of unmet demand.

Short and over, counted separately

The single figure above hides something. Not every line was under-supplied — some were over-supplied.

	Units
Short on lines that were under-supplied	5,04,052
Extra on lines that were over-supplied	35,867

	Units
Net difference	4,68,431

These two numbers differ by about 7.6%, and we report them separately rather than netting them off. The reason is practical: if a branch is short 100 units of a windscreen and over-supplied 100 units of a door glass, the net is zero but the customer still could not be served. Netting them off would make the service problem disappear on paper.

Service is getting worse, and clearly so

This is the most significant thing in the data.

Month	Ordered	Despatched	Served
Apr 2025	1,38,742	1,20,362	86.8%
May 2025	1,66,377	1,47,678	88.8%
Jun 2025	1,63,842	1,45,115	88.6%
Jul 2025	1,85,933	1,62,717	87.5%
Aug 2025	1,37,385	1,22,036	88.8%
Sep 2025	1,59,102	1,34,105	84.3%
Oct 2025	1,47,210	1,23,667	84.0%
Nov 2025	1,53,704	1,36,204	88.6%
Dec 2025	1,56,341	1,37,987	88.3%
Jan 2026	1,48,140	1,33,132	89.9%
Feb 2026	1,49,256	1,29,985	87.1%
Mar 2026	1,56,754	1,18,291	75.5%
Apr 2026	1,84,634	1,41,048	76.4%
May 2026	1,60,937	1,19,170	74.0%
Jun 2026	1,92,956	1,23,560	64.0%
Jul 2026	2,01,079	1,38,904	69.1%

For the first eleven months service sat between 84% and 90%. From March 2026 it drops sharply and does not recover — **June 2026 is the worst month in the file at 64%.**

What makes this more pointed is that **demand did not fall while this was happening — it rose**. June and July 2026 are the two largest ordering months in the entire sixteen months. So the gap is not branches asking for less; it is the business supplying a smaller share of a growing order book. That is a supply-side problem, and it is worth AIS looking at directly.

Demand is concentrated

	Share of all units ordered
Top 10 products	19.4%
Top 50 products	44.5%
Top 100 products	57.0%
Top 500 products	87.4%

A hundred products out of 2,063 account for well over half the volume. The remaining 1,563 products share about one-eighth of it between them. The same pattern holds across branches: the top 20 of 53 branches account for 61.3% of units.

The biggest branches

Branch	Units ordered	Share	Served
Bengaluru	1,56,563	6.0%	81.6%
Jaipur	1,27,066	4.9%	82.4%
Ahmedabad	1,25,966	4.8%	84.0%
Secundrabad	1,15,379	4.4%	82.2%
Karnal	90,291	3.5%	77.0%
Delhi-1	81,579	3.1%	84.1%
Noida	80,185	3.1%	77.5%
Ludhiana	78,159	3.0%	73.0%
Pune	74,191	2.9%	90.1%
Surat	69,393	2.7%	88.9%

Service varies a great deal by branch

The 82% network figure hides a wide spread. Among branches ordering more than 5,000 units:

Best served		Worst served	
Bhubaneswar	94.5%	Delhi-E1	66.5%
Koperkhane	94.1%	Jammu	70.6%
Andheri	94.0%	Chandigarh	71.4%
Kolkatta	90.5%	Ludhiana	73.0%
Pune	90.1%	Jalandhar	73.4%
Madurai	89.6%	Cochin	74.0%

There is a 28-point gap between the best and worst served branches. Four of the six worst are in the north — Delhi-E1, Jammu, Chandigarh, Ludhiana and Jalandhar — which suggests the problem may be regional rather than branch-by-branch.

What we predict

For the pilot we take two branches, Bengaluru and Delhi-1, and 136 products — 261 branch-and-product combinations in total. For each of those we estimate how many units will be needed in the coming months, and from that we work out how much should be ordered now, allowing for what is already in stock, what is already on its way, and how long that branch takes to receive goods.

We predict what branches will **ask for**, not what they were sent. This distinction matters. If we predicted despatches, we would be predicting our own supply failures and building them into next year's plan — a branch that was starved last year would be assumed to need less this year. Predicting the request keeps the real demand visible.

How we checked

The way we checked matters, because the easy way of checking proves nothing.

If we simply read the figures our own system had already worked out and compared them against our own screens, all we would prove is that our screens agree with our own calculations. If a rule was wrong somewhere, both sides would be wrong in exactly the same way and the check would sail through.

So instead we **went back to the client's original files and worked every figure out again from scratch**, using a completely separate piece of working that shares nothing with the system. It opens the spreadsheets, adds up the columns itself, and knows nothing about how the system stores or calculates anything. Only then did we put the two sets of numbers side by side.

Making sure it was the same data

Before comparing anything, we checked that the files sitting in our system were the identical files AIS sent, not a newer or partly-updated copy. Every file was compared against the original download, byte for byte. All five are identical:

File	Size in bytes	Identical?
Location Master	24,477	Yes
Orders & Receipts (Lead Time)	6,75,71,606	Yes
Sales Data FY 24~26	8,24,29,868	Yes
Stock in Hand as on 1st Aug'26	69,95,927	Yes
Substitution Mapping	2,23,570	Yes

This rules out the most ordinary explanation for any difference — that the two sides were simply looking at different copies of the file.

Matching the periods

Every comparison covers the same stretch of time: **1 April 2025 to 31 July 2026**, which is the full span of the order history. Our system also holds earlier data drawn from the sales files, so comparing an unfiltered total against the order file would have made our side look larger for no good reason. Every figure in this report is restricted to the period the order file actually covers.

What we compared

Not just the grand totals. A grand total can come out right while the detail underneath is wrong in two places that happen to cancel each other out. So we checked every breakdown separately: all 53 branches, every glass type, every importance class, every vehicle category, the largest products by volume, and every single month of the trend.

The main result

Across the whole business — 53 branches, 2,063 products

	In the AIS files	In our system	Result
Order lines read	7,75,912	7,75,912	Identical
Units ordered	26,02,392	26,02,392	Identical
Units despatched	21,33,961	21,33,961	Identical
Units short	5,04,052	5,04,052	Identical
Served	82.00%	82.00%	Identical
Order value	₹8,57,12,23,655	₹8,57,10,57,107	₹1,66,548 apart
Number of branches	53	53	Identical
Number of products	2,063	2,063	Identical
Period covered	1 Apr 25 – 31 Jul 26	1 Apr 25 – 31 Jul 26	Identical

These are not rounded figures picked to look neat. Twenty-six lakh two thousand three hundred and ninety-two units on both sides, arrived at twice, independently, from the same file.

Across the pilot — Bengaluru and Delhi-1, 136 products

	In the AIS files	In our system	Result
Units ordered	1,46,814	1,46,814	Identical
Units despatched	1,29,559	1,29,559	Identical
Units short	20,449	20,449	Identical
Served	88.25%	88.25%	Identical
Order value	₹59.95 crore	₹59.95 crore	Identical

On the pilot every figure matches exactly, rupee value included. The two order lines responsible for the difference at network level both happen to sit outside these two branches.

About that ₹1,66,548

It works out at 0.019% of total order value, and it comes from exactly **two order lines pulling in opposite directions**. Both are set out in full further on. Apart from those two, **51 of the 53 branches match to the rupee**, and all 53 match exactly on every quantity.

Every chart, checked one at a time

Each chart on the screens was checked on its own against the original files.

By branch — all 53 of them

Every single branch matches exactly on units ordered and units despatched. No exceptions and no tolerance allowed. On rupee value, 51 of the 53 match exactly; Jabalpur and Coimbatore are the two covered in the next section.

By glass type

Glass type	AIS files	Our system
Laminated	16,74,670	16,74,670
Sidelite	5,77,327	5,77,327
Backlite	3,47,748	3,47,748
Not in product list	2,641	2,641
Tempered windscreen	6	6
Total	26,02,392	26,02,392

By importance class

Class	AIS files	Our system
A	19,19,460	19,19,460
B	4,40,196	4,40,196
C	1,73,231	1,73,231
New Model	61,085	61,085
D	5,779	5,779
Not in product list	2,641	2,641

Class	AIS files	Our system
Total	26,02,392	26,02,392

Class A products are 19.2 lakh of the 26 lakh units — just under three-quarters of everything ordered.

By vehicle category

Category	AIS files	Our system
Car & MUV	22,34,409	22,34,409
Commercial	2,57,626	2,57,626
Three-wheeler	1,04,998	1,04,998
High End	2,718	2,718
Not in product list	2,641	2,641
Total	26,02,392	26,02,392

One thing came out of this check that is worth passing on. The product list spells this category **two different ways** — "Car & MUV" against 1,530 products and "Car & Muv" against another 539. Taken literally those are two separate categories, and the chart would split one real category into two bars with no warning. We treat them as the same thing, which is why the figures above line up. The same applies to "Commercial" and "High End". AIS may want to tidy this in the source file.

By product

All 2,063 products appear on both sides, and the ten largest match exactly, in the same order:

Product code	AIS files	Our system
FG.MP8.LFH.GCG2120000	92,636	92,636
FG.MM7.LFH.GCGB1B0000	64,745	64,745
FG.TCF.LFH.GCG2120000	61,653	61,653
FG.BA5.LFH.GCG2120000	61,614	61,614
FG.MYA.LFH.GCG2120000	41,246	41,246

Product code	AIS files	Our system
FG.MPH.LFH.GCG21X0000	39,890	39,890
FG.MY7.LFH.GCG2120000	39,647	39,647
FG.M1C.LFH.GCG2120000	34,862	34,862
FG.MYS.LFH.GCG2120000	34,847	34,847
FG.ME3.LFH.GCG2120000	34,126	34,126

The monthly trend

All sixteen months match exactly, both on units and on the percentage served.

This one needed some care. The data is held week by week, and a week that falls across the end of a month has to be counted in one month or the other. The convention is that the week belongs to whichever month holds its Thursday. Checked that way, all sixteen months agree exactly. Checked against plain calendar months, every month disagrees — which is a difference in how the weeks are grouped, not a difference in the data. More on that in the next section but one.

The two order lines that do not agree

The entire ₹1,66,548 comes down to two lines out of 7,75,912. We have traced both to the individual order. They pull in opposite directions, which is why the net figure is smaller than either one on its own.

Jabalpur — one line where the file contradicts itself

Branch	Jabalpur
Order number	E1/05-25/02/JLR
Order date	1 May 2025
Product	FG.VL1.LFH.GCG3130000
Quantity on the line	22
Rate on the line	₹21,500
Value written in the file	₹7,09,500
Quantity × rate	₹4,73,000

Gap	₹2,36,500
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The value written in the file, ₹7,09,500, is exactly $33 \times ₹21,500$. But the quantity column on the same line says 22. The two columns in the client's own file disagree with each other, and there is nothing in the file that tells us which one is right.

We work value out as quantity $\times$ rate, so our Jabalpur figure comes out **₹2,36,500 lower** than the value written in the file. We have deliberately not adjusted it to match, because to do that we would have to override a stated quantity with a figure we cannot verify.

This is the **only line in all 7,75,912** where the file's own value and its own quantity $\times$ rate disagree.

Coimbatore — 44 lines with no price on them

44 order lines at Coimbatore, covering 246 units between them, have no rate recorded at all. The value written against them in the file is zero.

We price those units at the going rate for the same product at the same branch in the same week. That adds **₹69,952**, so our Coimbatore figure comes out **higher** than the file's.

The thinking is straightforward: a unit that was genuinely ordered is genuinely demand, and pricing it at what its neighbouring lines sold for is more defensible than recording it as worth nothing at all. Recording it as zero would be just as much of an assumption — only a less visible one, because it would never show up as a difference.

It is worth noting that these same 44 lines also have no despatch quantity recorded. Both gaps point the same way: these look like records that were never completed, rather than real orders worth nothing.

The two together

	Effect on our figure
Jabalpur, 1 line	-₹2,36,500
Coimbatore, 44 lines	+₹69,952
Net	-₹1,66,548

Against a total order value of ₹857 crore that is 0.019%. Neither of them affects a single quantity anywhere, and neither falls inside the two pilot branches.

Three mistakes we found in our own working

The check did not pass first time. Three real mistakes turned up on our side. All three were silent — none of them produced an error or made a screen look obviously wrong, which is exactly what makes this kind of checking worth doing.

One — we were averaging prices the wrong way

Where a branch ordered the same product more than once in a week at different prices, we were treating every one of those prices as equally important. An order for 2 pieces counted just as much as an order for 2,000 pieces when working out the average price.

Every rupee figure on every screen is worked out as quantity × average price. So wherever more than one price was involved, that figure disagreed with the sum of the actual lines. **19,359 of 5,82,324 groupings had more than one price in them.** The total order value was coming out **₹2.92 crore too high.**

The fix is to weight the average by how many units were sold at each price, which is what anyone would do by hand. Value now adds back to the line-by-line total exactly.

Two — we were reading a blank despatch as a zero

When the despatch column on a line is empty, that means nobody recorded what was sent. We were treating it as nothing sent, which turned the entire ordered quantity into a shortfall.

44 lines are affected. Between them they were creating **246 units of shortfall that never happened** and dragging the service percentage down with it.

Blank and zero are now treated as two different things. A blank is recorded as unknown and left out of the service calculation, instead of being scored as a complete failure to supply.

Three — we were dropping products that were missing from the product list

Two products that AIS orders **across 44 of its 53 branches** do not appear anywhere in the Substitution Mapping product list. With no glass type, no class and no vehicle category against them, our charts were simply throwing those rows away.

The result was a chart that did not add up to its own heading. Every product breakdown came out **2,641 units short** of the total shown immediately above it, and there was nothing on the screen to explain where the difference had gone. Anyone adding up the bars would have found the gap and had no way to account for it.

Those rows now appear under a clearly marked heading, **"Not in product list"**, so they are visible, they are counted, and the breakdown adds back to the total. We named it after the actual reason rather than burying it under "Other", because it is a gap in the product list that AIS will probably want to close.

Why they will not come back

Each fix is held in place by an automatic check that tests the thing that was broken — for instance, that a product breakdown adds up to its own heading figure. One of those checks is written so that it fails if anyone ever puts the old behaviour back. There are 1,361 such checks across the system and all of them currently pass.

Two things that looked wrong but were not

These are worth recording, because both looked serious for a while and neither turned out to be a problem with the system. In both cases the checking was at fault, not the data.

The branch comparison that showed 60% of branches wrong

The first time we compared branch by branch, roughly three branches in five came out different, some of them badly so. That looked like a serious problem.

It was not. The comparison had been set up without the date filter on one side, so a sixteen-month total from the order file was being measured against a twenty-eight-month total that included the older sales history. Once both sides covered the same period, all 53 branches matched exactly.

The monthly trend where all sixteen months were wrong

The first monthly comparison showed every one of the sixteen months disagreeing. Again this looked bad, and again it was the comparison rather than the system.

The data is held week by week. A week that straddles the end of a month has to be counted in one month or the other, and the convention is that the week goes to whichever month contains its Thursday. The comparison had been built using plain calendar months instead. Once the correct rule was applied to both sides, all sixteen months matched exactly.

Why we are including this

Two reasons. First, honesty — the checking was not clean from the start and saying so is more useful than presenting a tidy story. Second, and more usefully: a number disagreeing does not by itself mean the system is wrong. Both of these would have been reported as

serious failures if we had stopped at the first result. Each one needed tracing to its cause before it meant anything, and in both cases the cause was in how we were measuring.

Where this leaves us

The system reproduces the client's data. Every quantity — ordered, despatched, short, and the percentage served — matches exactly at the overall level, at each of the 53 branches, across every product breakdown and across every month. On rupee value, 51 of 53 branches match to the rupee, and the remaining ₹1,66,548 comes from two order lines we can name.

That is as close to a complete match as this data allows, and the two exceptions are both gaps in the source file rather than errors in the calculation.

Five things to put to AIS

These all came out of the checking. None of them stops the work, but all of them are worth a conversation.

1. **Service has fallen sharply since March 2026 and has not recovered.** It ran at 84–90% for eleven months, then dropped to 75.5%, 76.4%, 74.0%, 64.0% and 69.1%. Orders went up over the same period, so this is not falling demand. June and July 2026 were the two biggest ordering months in the file and two of the three worst-served. This is the most important thing in the data and it sits outside the scope of what we were asked to check.
2. **Forty-four order lines at Coimbatore have no price and no despatch quantity.** Is there another source for these figures, or should the 246 units be treated as incomplete records? At present we price them at the going rate for neighbouring lines, which adds ₹69,952.
3. **One line at Jabalpur states a value its own quantity does not support.** Order E1/05-25/02/JLR shows 22 units at ₹21,500 but a value of ₹7,09,500, which is 33 units' worth. Which of the two columns is correct?
4. **Two products ordered across 44 branches are missing from the product list.** They account for 2,641 units over the period. Should they be added, or are they deliberately kept outside it?
5. **The vehicle category is spelt two ways in the product list.** "Car & MUV" on 1,530 products and "Car & Muv" on 539. We treat them as one category. Worth correcting at source so that anyone else reading the file gets the same answer we do.

What a full rollout would need

- **Stock history instead of a single date.** We have one photograph of stock, taken on 1 August 2026. Today's recommendations are sound, but without a record of stock over time there is no way to test how a stocking policy would have performed in the past.
- **A goods-received date.** The order file records when an order was placed and when it was despatched, but not when it actually arrived at the branch. So what we can currently measure is one leg of the journey, not the whole of it.
- **A decision on how much to allow for unreliable delivery times.** Some branches receive goods far less predictably than others — North 24 Parganas averages three days but swings by more than two days either side. At present it is given the same cover as a branch that delivers reliably in three days. Changing that would alter every recommended order quantity, so it is a business decision rather than something to adjust quietly.
- **Widening beyond the pilot.** The forecasting currently runs on two branches and 136 products. The whole network of 53 branches and 2,063 products is already loaded and checked, so extending it is a question of processing time rather than of approach.